

Experiment - 2

TestDisk: Open-Source Data Recovery Tool

recover a missing partition and repair a corrupted one using test disk

Installation

Linux : `sudo apt-get install testdisk`

macOS : `brew install testdisk`

Windows: Download the executable from the official website.

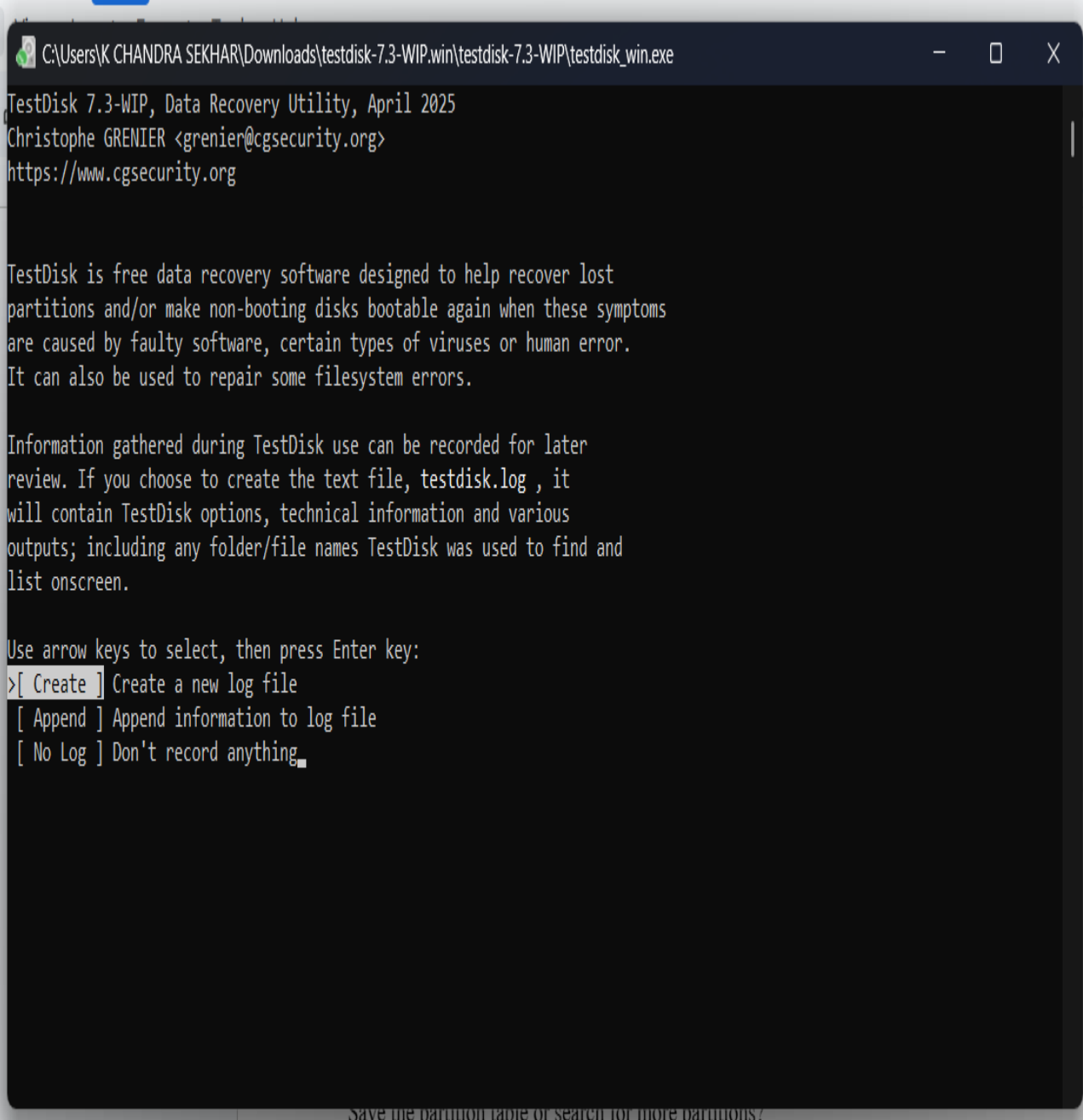
Link : https://www.cgsecurity.org/wiki/TestDisk_Download

NOTE:- if you using in windows please extract the .zip file first

Procedure

1. Create a Log File

- Launch TestDisk from your terminal or command prompt using `sudo testdisk` (or `testdisk_win.exe` on Windows).
- Select the [Create] option to generate a log file of the recovery session. This is helpful for future reference or troubleshooting.



```
C:\Users\K CHANDRA SEKCHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe

TestDisk 7.3-WIP, Data Recovery Utility, April 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

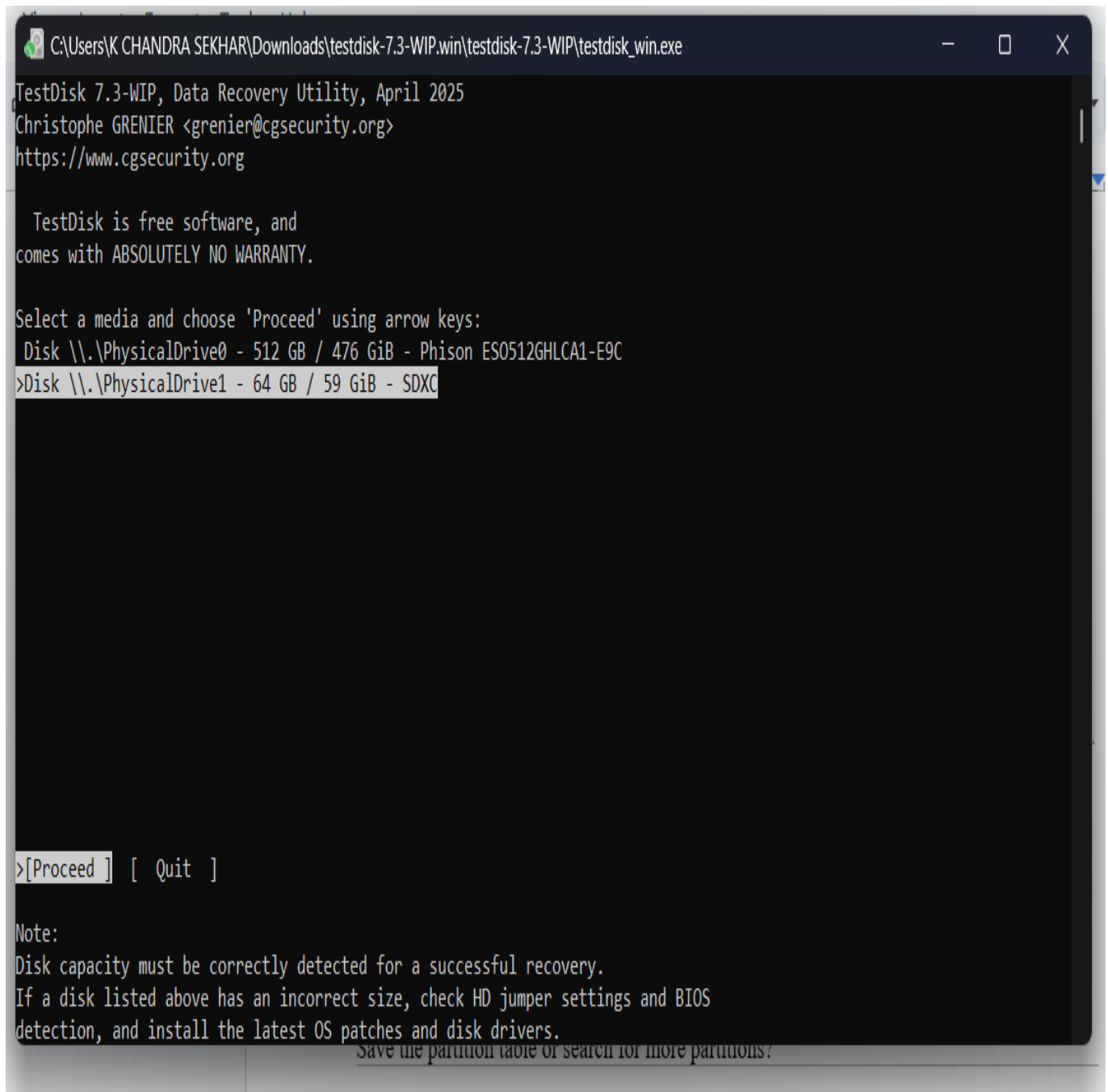
TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
>[ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything.
```

2. Select the Drive to Analyze

- TestDisk will display a list of all detected storage devices.
- Use the Up/Down arrow keys to highlight the drive that contains your lost data.
- Select [Proceed] to move to the next step.
-



```
C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe

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TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

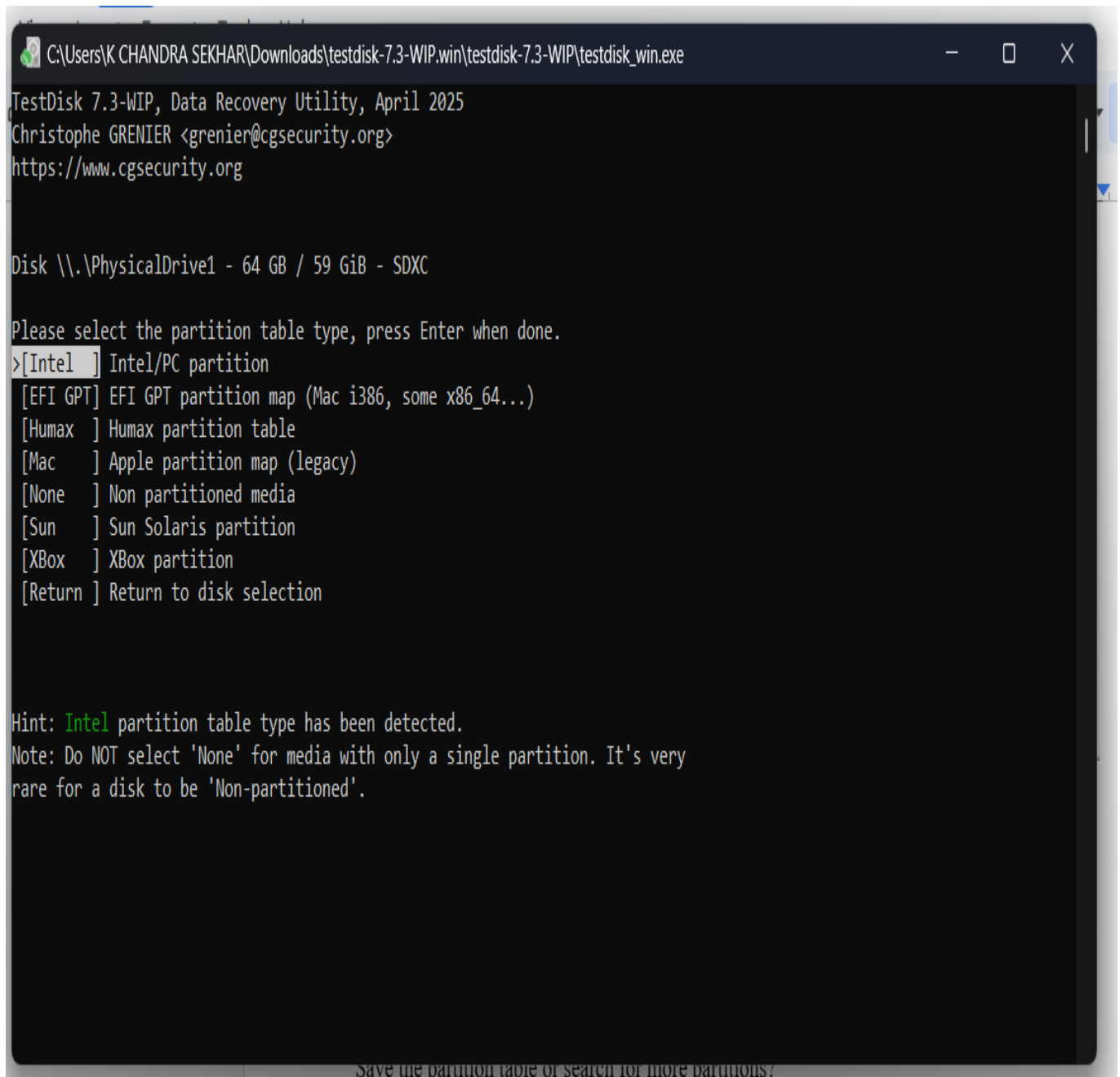
Select a media and choose 'Proceed' using arrow keys:
Disk \\.\\PhysicalDrive0 - 512 GB / 476 GiB - Phison ES0512GHLCA1-E9C
Disk \\.\\PhysicalDrive1 - 64 GB / 59 GiB - SDXC

>[Proceed] [ Quit ]

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
Save the partition table or search for more partitions?
```

3. Choose the Partition Table Type

- TestDisk will automatically suggest the most likely partition table type (e.g., Intel/PC, EFI GPT).
- The default value is usually correct. Confirm the selection by pressing Enter.



```
C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, April 2025
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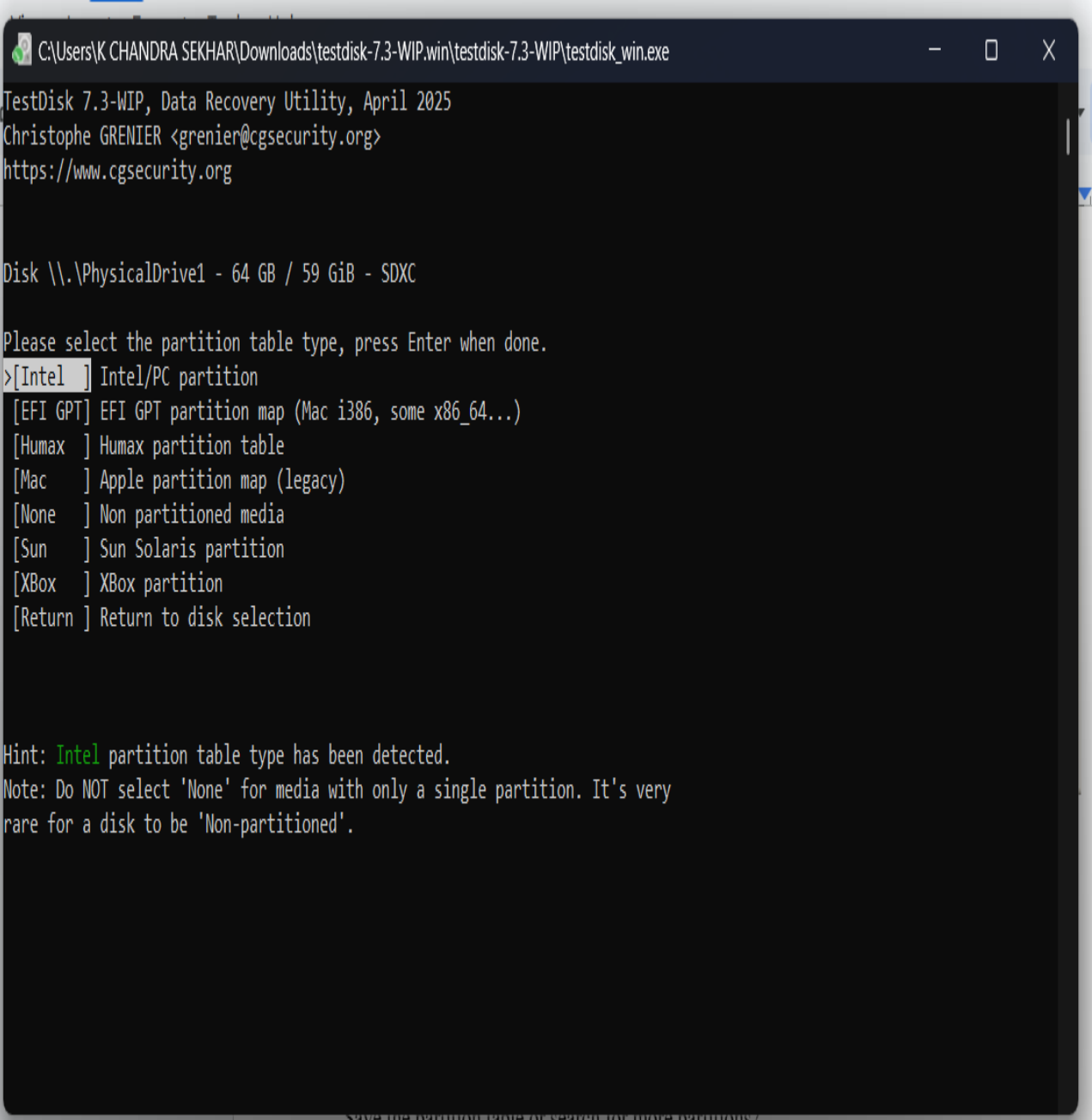
Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - SDXC

Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax  ] Humax partition table
[Mac    ] Apple partition map (legacy)
[None   ] Non partitioned media
[Sun    ] Sun Solaris partition
[XBox   ] Xbox partition
[Return ] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

4. Analyze Current Partition Structure

- From the main menu, choose [Analyse] and press Enter.
- This will display the current partition table and check for errors or missing partitions.



```
C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - SDXC

Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax  ] Humax partition table
[Mac    ] Apple partition map (legacy)
[None   ] Non partitioned media
[Sun    ] Sun Solaris partition
[XBox   ] Xbox partition
[Return ] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.

Save the partition table or search for more partitions?
```

5. Perform a Quick Search

- After the analysis, you will be prompted to perform a [Quick Search].
- Select it and press Enter to scan the drive for lost partitions.

The screenshot shows the TestDisk 7.3-WIP interface. The title bar indicates the file path: C:\Users\K CHANDRA SEKHAAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe. The main window displays the following information:

```

TestDisk 7.3-WIP, Data Recovery Utility, April 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63
Current partition structure:
    Partition          Start      End    Size in sectors

  1 P HPFS - NTFS      2 10 9 7791 180 5 125140992
No partition is bootable

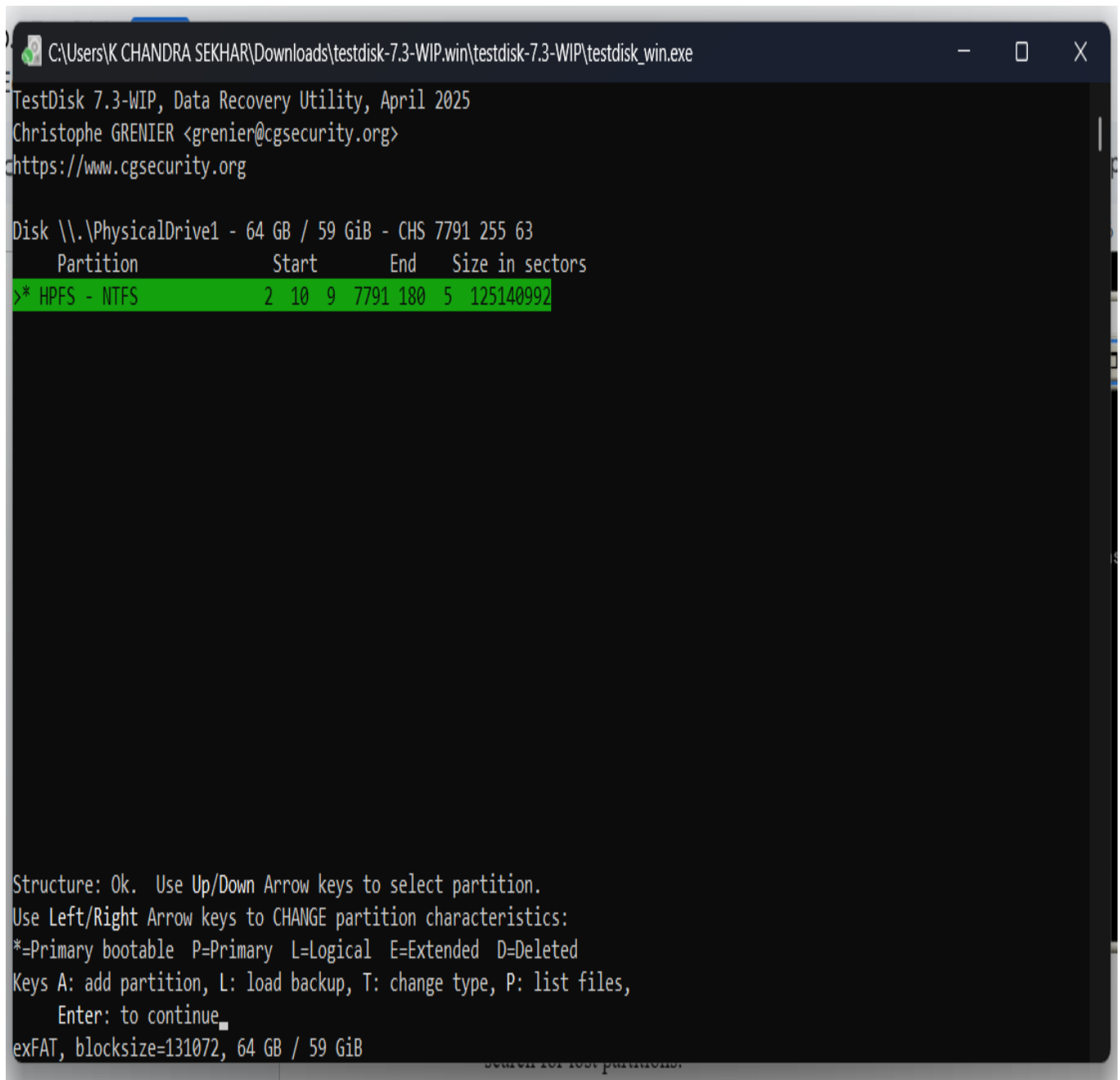
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
>[Quick Search] [ Backup ]
Try to locate partition.
Save the partition table or search for more partitions?

```

- Once a partition is found, you can press p to list its files. Deleted files and folders often appear in red.
- Press q to return to the search results.

6. Perform a Deeper Search

- If the Quick Search fails to find your lost partitions, select [Deeper Search]
- This process can take a long time, as it scans the entire drive, block by block, to find remnants of partition structures.



The screenshot shows the TestDisk 7.3-WIP interface. At the top, it displays the file path: C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe. Below this, it shows the version and date: TestDisk 7.3-WIP, Data Recovery Utility, April 2025. The developer's name and email are listed: Christophe GRENIER <grenier@cgsecurity.org>, along with the website: https://www.cgsecurity.org.

The disk information is shown as: Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63. Below this, a table lists the partitions:

Partition	Start	End	Size in sectors
* HPFS - NTFS	2 10 9	7791 180 5	125140992

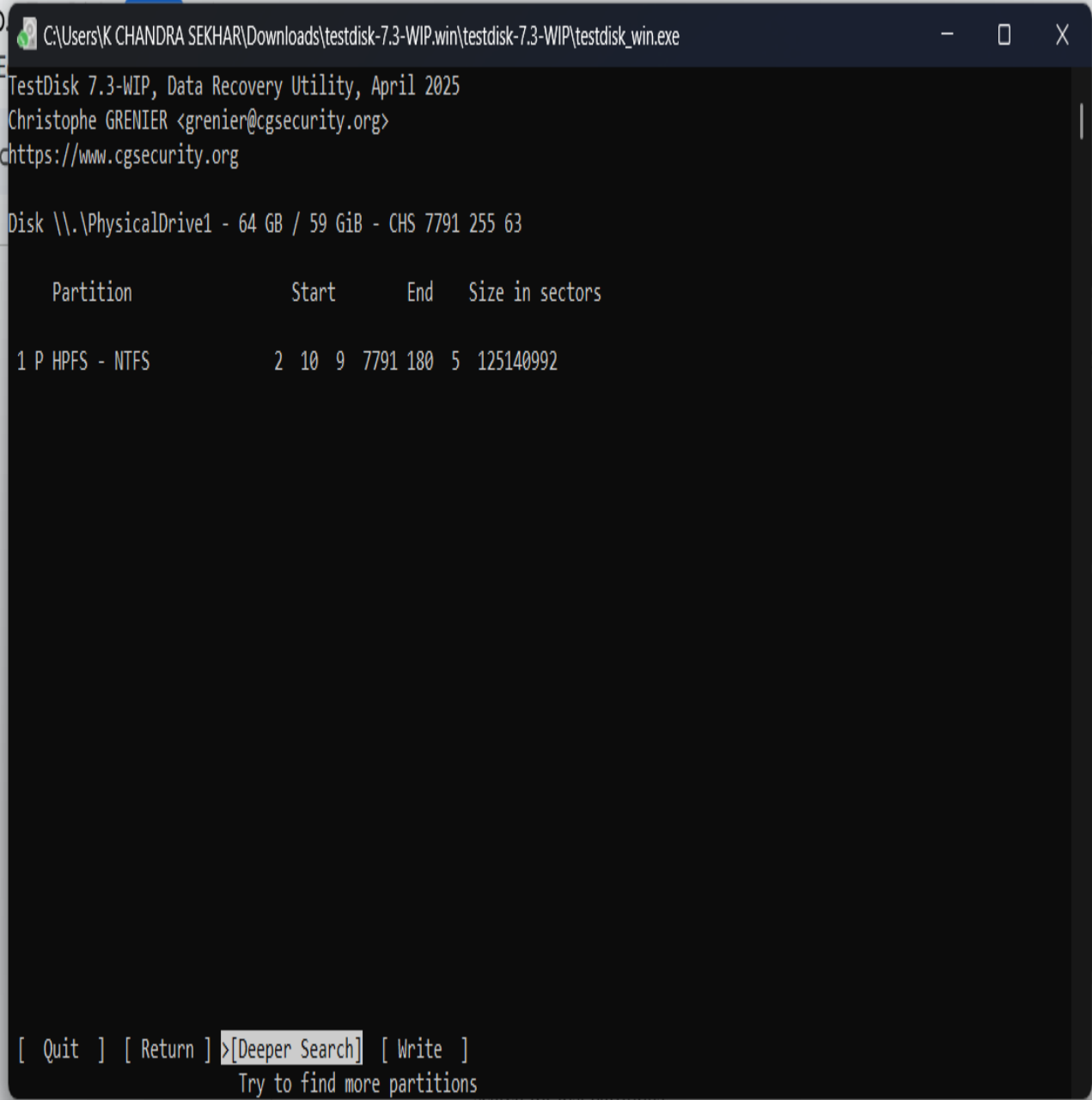
Below the table, it says: Structure: Ok. Use Up/Down Arrow keys to select partition. Use Left/Right Arrow keys to CHANGE partition characteristics: *=Primary bootable P=Primary L=Logical E=Extended D=Deleted. Keys A: add partition, L: load backup, T: change type, P: list files, Enter: to continue. At the bottom, it shows: exFAT, blocksize=131072, 64 GB / 59 GiB.

- Again, use p to preview files and confirm if a found partition is the one you are looking for.

7. Modify Partition Status

- After finding the correct partitions, use the Left/Right arrow keys to set their status.

- Use **Left/Right arrow keys** to change status:



The screenshot shows the TestDisk 7.3-WIP interface. At the top, it displays the title bar with the file path: C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe. Below the title bar, the text reads: "TestDisk 7.3-WIP, Data Recovery Utility, April 2025", "Christophe GRENIER <grenier@cgsecurity.org>", and "https://www.cgsecurity.org". The disk information is shown as "Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63". A table of partitions is displayed with the following columns: Partition, Start, End, and Size in sectors. The table contains one entry: "1 P HPFS - NTFS" with Start "2 10 9", End "7791 180 5", and Size "125140992". At the bottom, a menu is visible with options: "[Quit]", "[Return]", ">[Deeper Search]", and "[Write]". Below the menu, it says "Try to find more partitions."

```
C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, April 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63

Partition                Start      End  Size in sectors
1 P HPFS - NTFS          2 10 9 7791 180 5 125140992

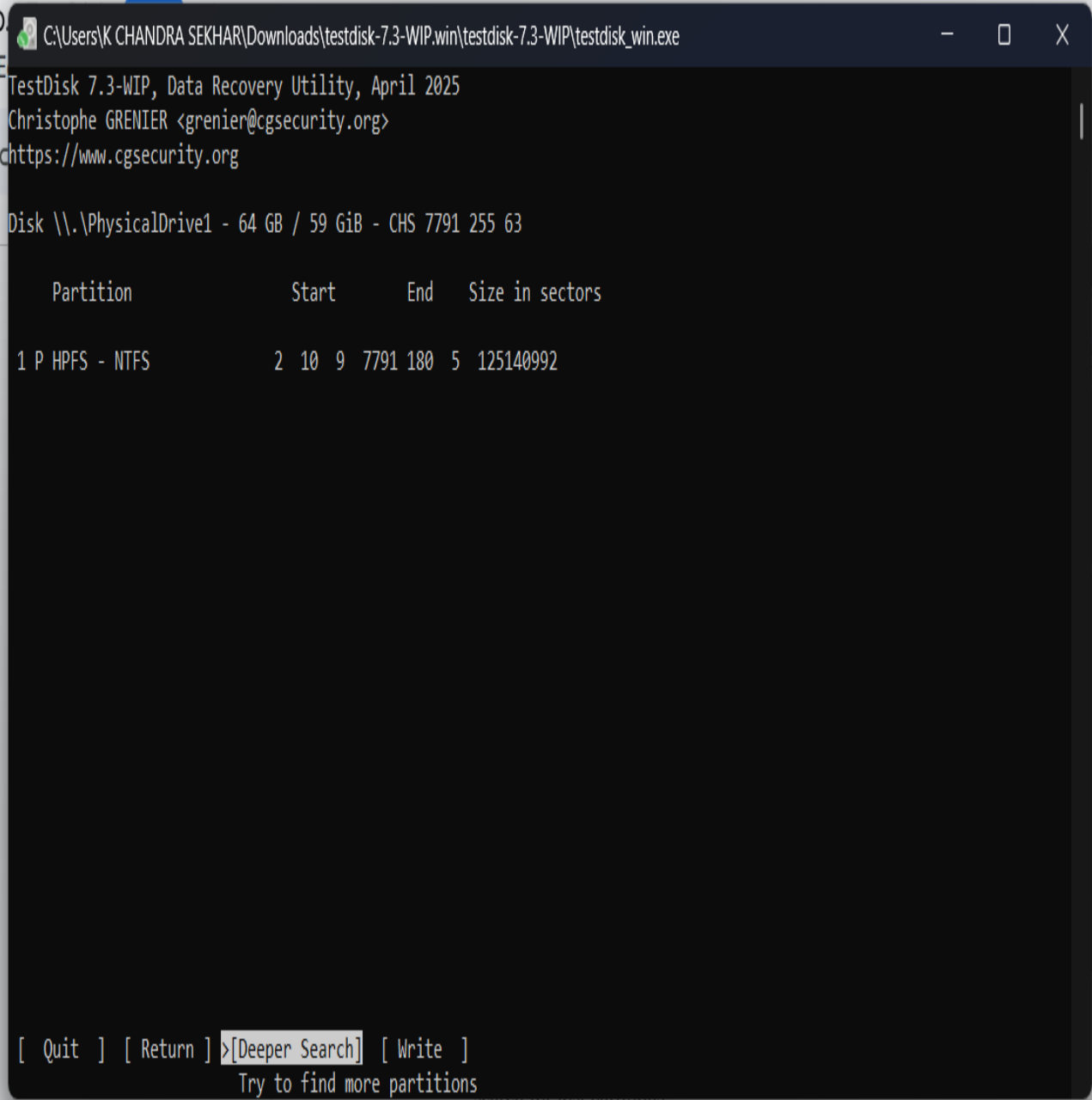
[ Quit ] [ Return ] >[Deeper Search] [ Write ]
Try to find more partitions.
```

- **P**: Primary
- *****: Bootable
- **L**: Logical
- **D**: Deleted
- ensure that the partitions you want to recover are marked as Primary or Logical (and not deleted).

8. Write the Partition Table

- Once you are confident the partition structure is correct, select [Write] from the menu.

- Confirm the operation by pressing y (for yes). This will write the new partition table to your disk.



The screenshot shows the TestDisk 7.3-WIP interface. At the top, it displays the title bar for the application. Below the title bar, it shows the version and author information. The main display area shows the disk information for Disk \\.\PhysicalDrive1, which is 64 GB / 59 GiB in size. Below this, a table lists the partitions. The table has four columns: Partition, Start, End, and Size in sectors. There is one partition listed: 1 P HPFS - NTFS, starting at 2 10 9 and ending at 7791 180 5, with a size of 125140992 sectors. At the bottom, there are navigation options: [Quit], [Return], >[Deeper Search], and [Write]. Below these options, it says 'Try to find more partitions'.

```
C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, April 2025
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63

Partition          Start      End      Size in sectors
1 P HPFS - NTFS    2 10 9 7791 180 5 125140992

[ Quit ] [ Return ] >[Deeper Search] [ Write ]
Try to find more partitions
```

```
4. Select the Partition Table Type

C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe

TestDisk 7.3-WIP, Data Recovery Utility, April 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - SDXC
CHS 7791 255 63 - sector size=512

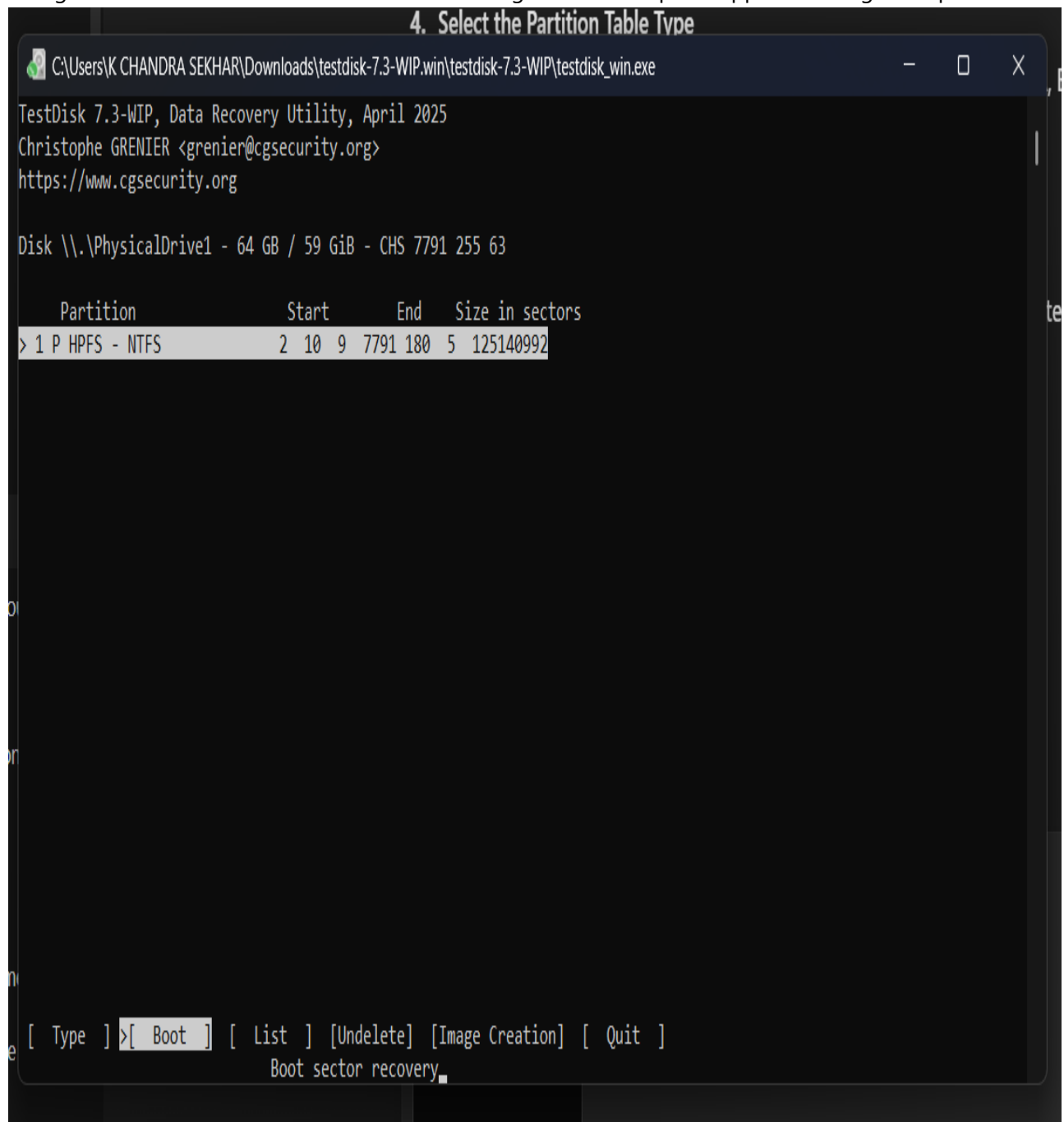
[ Analyse ] Analyse current partition structure and search for lost partitions
>[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete ] Delete all data in the partition table
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

9. Recover Files

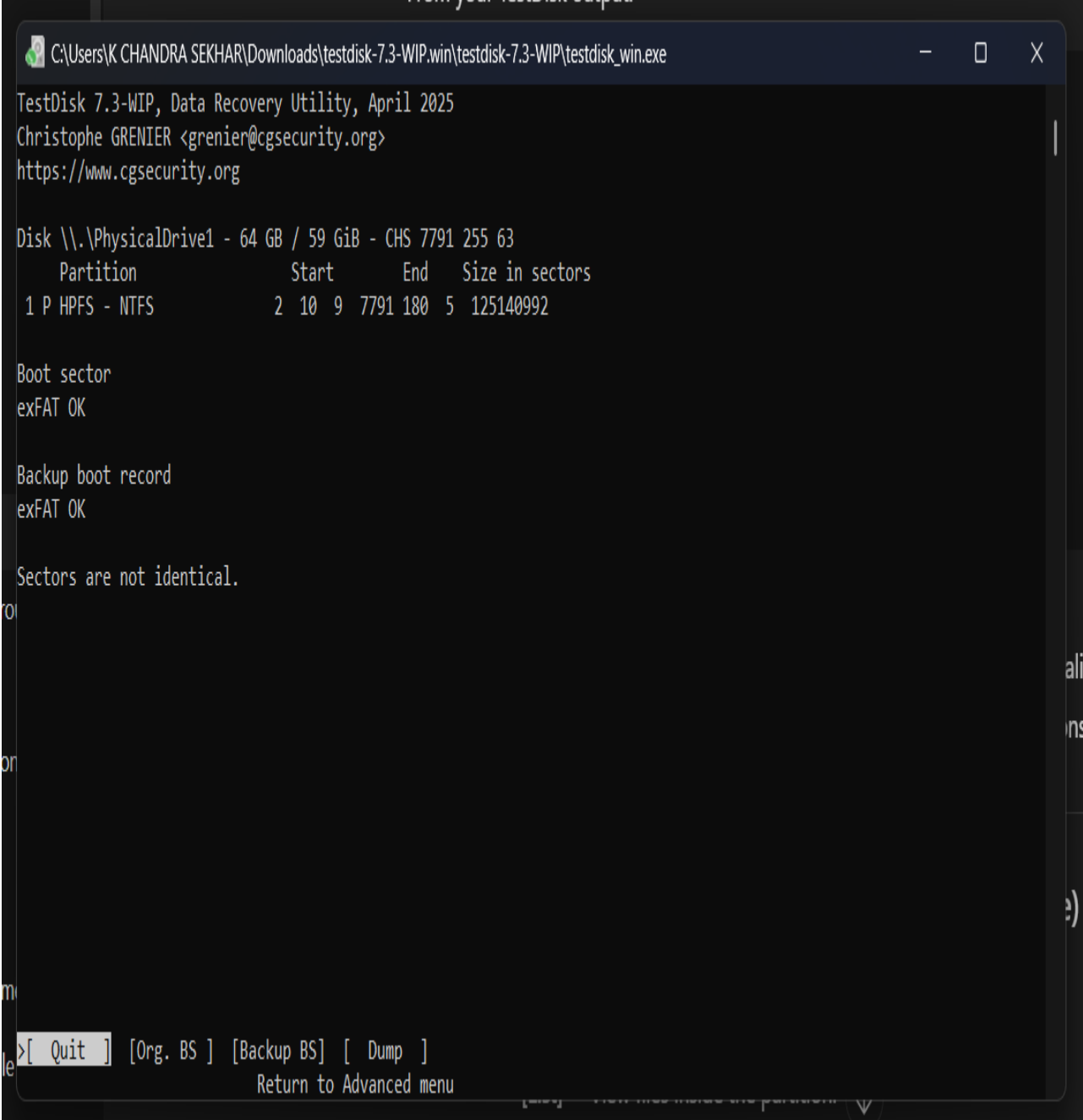
- If you just need to recover a few files without fixing the partition table, you can do so from the file list (after pressing p).
- Navigate to the folder containing your desired files.
- Use the colon : key to select the files you want to recover.
- Press the uppercase C key to copy the selected file(s).

- Navigate to a safe destination on a different storage device and press uppercase C again to paste.



10. Exit and Restart

- Once your task is complete, select [Quit] to exit the program.
- If you wrote a new partition table to the drive, it is recommended to restart your computer to allow the operating system to recognize the changes.



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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63

Partition	Start	End	Size in sectors
1 P HPFS - NTFS	2 10 9	7791 180 5	125140992

Boot sector
exFAT OK

Backup boot record
exFAT OK

Sectors are not identical.

At the bottom, there is a menu with options: >[Quit] [Org. BS] [Backup BS] [Dump] and a prompt "Return to Advanced menu".

References:

https://www.cgsecurity.org/wiki/TestDisk_Download

https://www.cgsecurity.org/testdisk_doc/